Properties of Fourier Transform

1. Linearity

f(t)	F(ω)
$af_1(t)+bf_2(t)$	$aF_1(\omega)+bF_2(\omega)$

2. Time-Shifting

f(t)	F(ω)
$f(t-t_o)$	$F(\omega)e^{-j\omega t_o}$

3. Frequency-Shifting

f(t)	F(ω)
$f(t)e^{j\omega_o t}$	$F(\omega-\omega_o)$

4. Time Scaling

f(t)	F(ω)
$f(at)$	$\frac{1}{\lvert a\rvert}F\left(\frac{\omega}{a}\right)$

5. Time Reversal

f(t)	F(ω)
$f(-t)$	$F(-\omega)$

6. Duality/Similarity

f(t)	F(ω)
$F(t)$	$2\pi f(-\omega)$

7. Time Differentiation

f(t)	F(ω)
$f'(t)$	$j\omega F(\omega)$

8. Integration in the time domain

f(t)	F(ω)
$\int_{-\infty}^{\tau} f(\tau)d\tau$	$\pi F(0)\delta(0)+j\omega F(\omega)$

9. Convolution

f(t)	F(ω)
$x(t)*h(t)$	$X(\omega)*H(\omega)$

Note: The convolution in the time domain is equivalent to multiplication in the frequency domain.

z-Transform

Another type of a transform of a sequence closely related to the Fourier transform is called the z-transform. It is defined as:

$$F(z)=\sum_{n=-\infty}^{\infty} f(n)z^{-n}$$

This is also called the **bilateral z-transform**.

Another definition of z-transform is:

$$F(z)=\sum_{n=0}^{\infty} f(n)z^{-n}$$

This is called the **unilateral z-transform**.

z-Transform of some elementary functions

	f(t)	F(z)
1.	1	$\frac{z}{z-1}$
2.	t	$T\frac{z}{(z-1)^2}$
3.	t^2	$T^2\frac{z(z+1)}{(z-1)^3}$
4.	e^{-at}	$\frac{z}{z-e^{-at}}$
5.	te^{-at}	$T\frac{ze^{-at}}{(z-e^{-at})^2}$
6.	$\frac{1}{a}(1-e^{-at})$	$\frac{z(1-e^{-at})}{a(z-1)(z-e^{-at})}$
7.	$\sin(\omega_o t)$	$\frac{z\sin\omega_o t}{z^2-2z\cos\omega_o t+1}$
8.	$\cos(\omega_o t)$	$\frac{z(z-\cos\omega_o t)}{z^2-2z\cos\omega_o t+1}$

Properties of z-Transforms

1. Linearity

f(t)	F(z)
$af_1(t)+bf_2(t)$	$aF_1(z)+bF_2(z)$

2. Shift left by n

f(t)	F(z)
$f(t+n)$	$z^n\left(F(z)-\sum_{t=0}^{n-1} f(t)z^{-k}\right)$

3. Shift right by n

f(t)	F(z)
$f(t-n)$	$z^{-n}F(z)$

4. Multiplication by time

f(t)	F(z)
$tf(t)$	$-z\frac{dF(z)}{dz}$

5. Convolution

f(t)	F(z)
$f_1(t)*f_2(t)$	$F_1(z)*F_2(z)$

Power Series

A power series is an expression in the form of $f(x)=\sum_{n=0}^{\infty} a_n(x-x_o)^n$

where: x_o = base point (fixed number)
A power series with a base point $x_o = 0$ is called a Taylor series or a Maclaurin series. With every power series, there is a unique number r such that $0\le r\le+\infty$. This number is called the radius of convergence. The series converges for $\lvert x-x_o\rvert<r$ and diverges for $\lvert x-x_o\rvert>r$. (x_o-r, x_o+r) is called the interval or the radius of convergence.

Fourier Series

If a function is expanded as a power series $f(x)=\sum_{n=0}^{\infty} a_n(x-x_o)^n$, about some point x_o, with radius of convergence r > 0, it is being approximated through the interval (x_o-r, x_o+r), by the polynomials $a_o+a_1(x-x_o)+\ldots+a_n(x-x_o)^n$, but the approximation is much better near x_o than $x_o\pm r$. This fact makes power series the approximation of choice for values of x near x_o. On the other hand, representing a function as a Fourier series consists of approximating the function by trigonometric polynomials. The approximation of f(x) using the Fourier series if it is periodic with period of 2π is:

$$f(x)=\frac{a_o}{2}+\sum_{n=1}^{\infty} a_n\cos nx+b_n\sin nx$$

where:

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos(nx)dx$$

$$b_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin(nx)dx$$